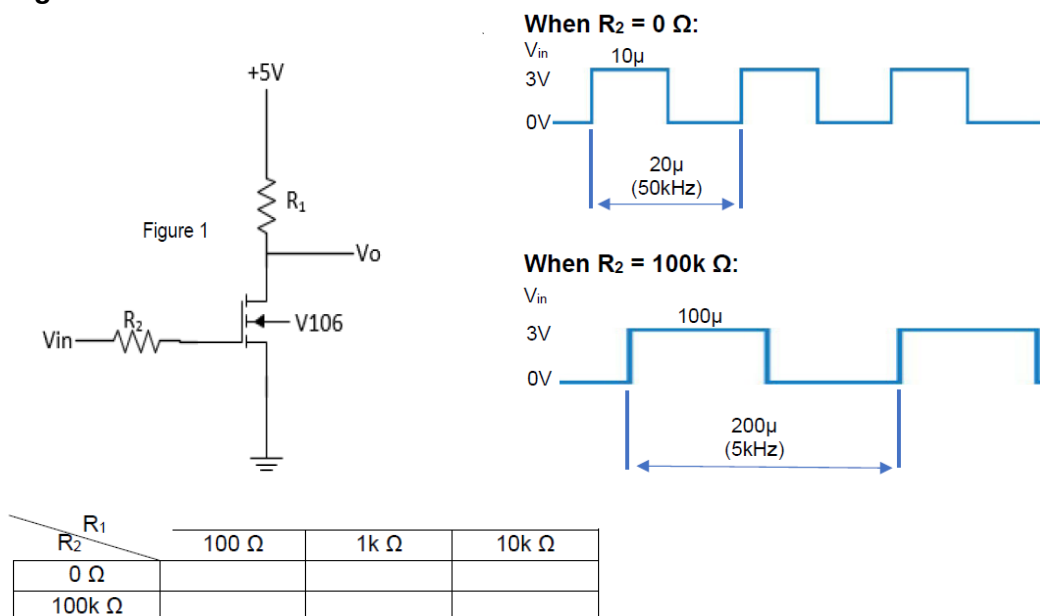


Experiment #9 Large Signal Characteristics of MOSFET

Executive Summary: In this experiment we tested out different resistances at the gate and the drain to find the time domain characteristics. That being, the Rise and fall times as well as time delay on the circuit across the MOSFET. $\tau=RC$ which was previously learned in the 2200 semester was explored and we were able to see how the MOSFET was affected by the time delay. This is due to every leg of the MOSFET creating a small capacitor and holding charge. Manipulation of the gate with smaller to larger resistance caused faster to slower response times respectively. Furthermore, in this lab we observed how the MOSFET is different as it is a VOLTAGE controlled current source and because of this we are allowed to change the resistance at the base all we want.

Objective: To study the large signal characteristics of metal-oxide-semiconductor field-effect transistors through time domain characteristics.

Diagrams:



Data:

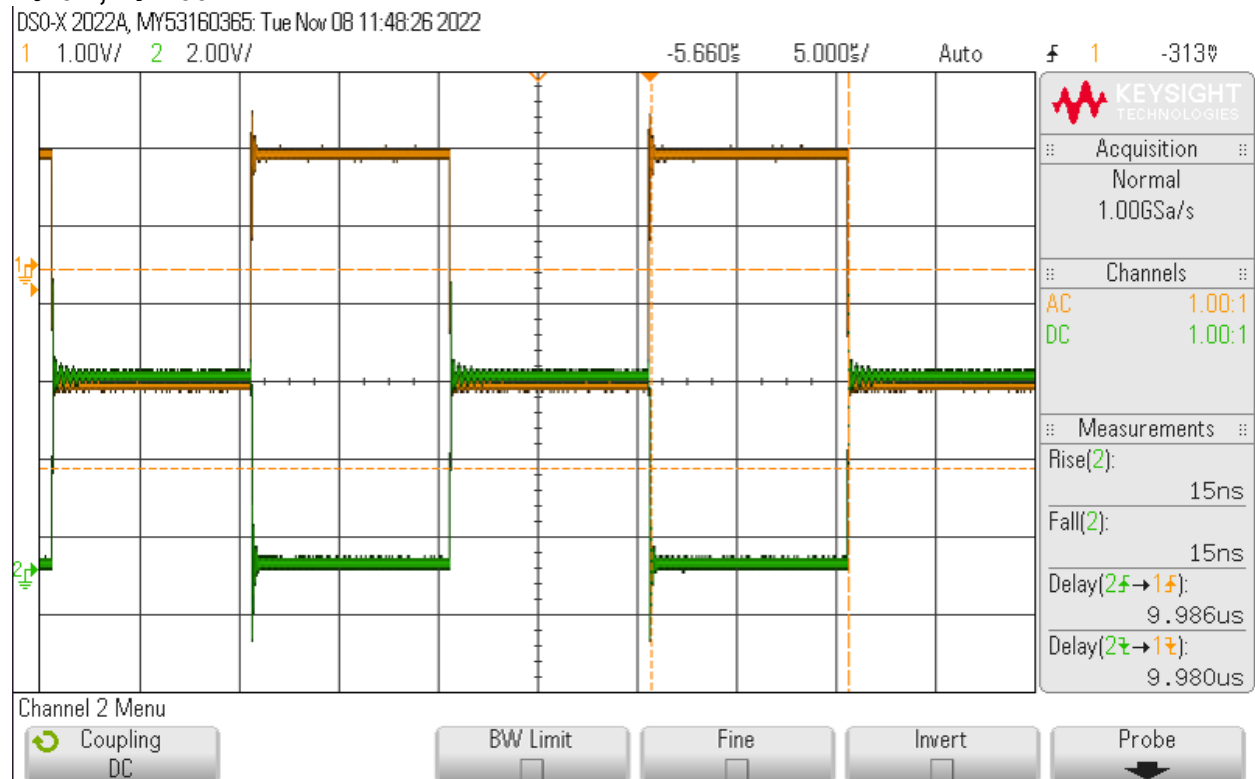
Component	Nominal Value	Measured Value	Percent Error(%)
Resistor 1	0Ω	0Ω	0
Resistor 2	100Ω	98.85Ω	1.2
Resistor 3	1kΩ	0.9907kΩ	0.9
Resistor 4	10kΩ	9.975kΩ	0.3
Resistor 5	100kΩ	100.22kΩ	0.2

Analysis:

In Part One, Figure 1 circuit was built and by following the table provided, we find R_o , R_s . The V_{in} supply wave former was set as instructed and while R_o was initially set to 0Ω, the oscilloscope measuring input and output we analyzed the rise time (t_r), fall time(t_f), time delay in rise (t_{dr}), and the time delay in fall(t_{df}) for each circuit. Part two consists of following the provided chart for the rest of the resistors, 1kΩ and 10kΩ and take the measurements respectively. In Part Three, again, following the chart we changed R_o to 100kΩ and finished the final resistors of 100Ω, 1kΩ and 10kΩ the wave former was set respectively. All measurements of resistors were recorded.

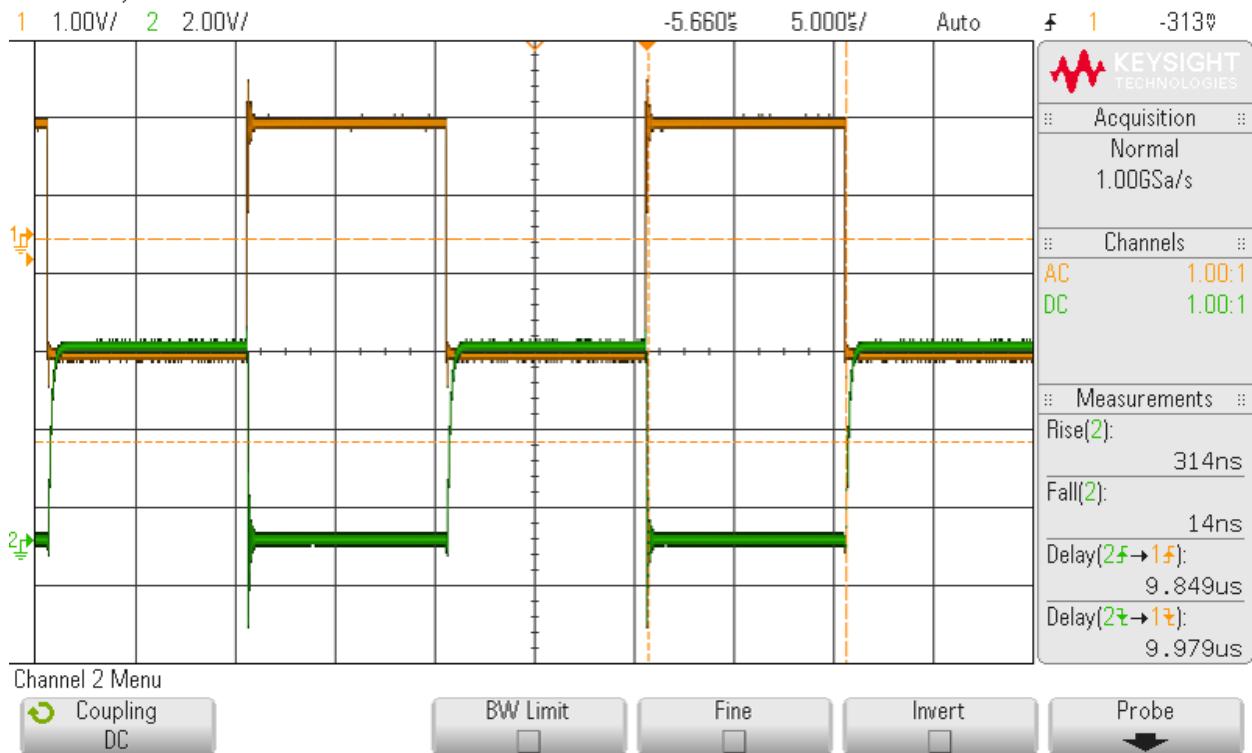
Note: Yellow waveform= input
Green waveform=output

$R_c=0\Omega$, $R_o=100\Omega$:



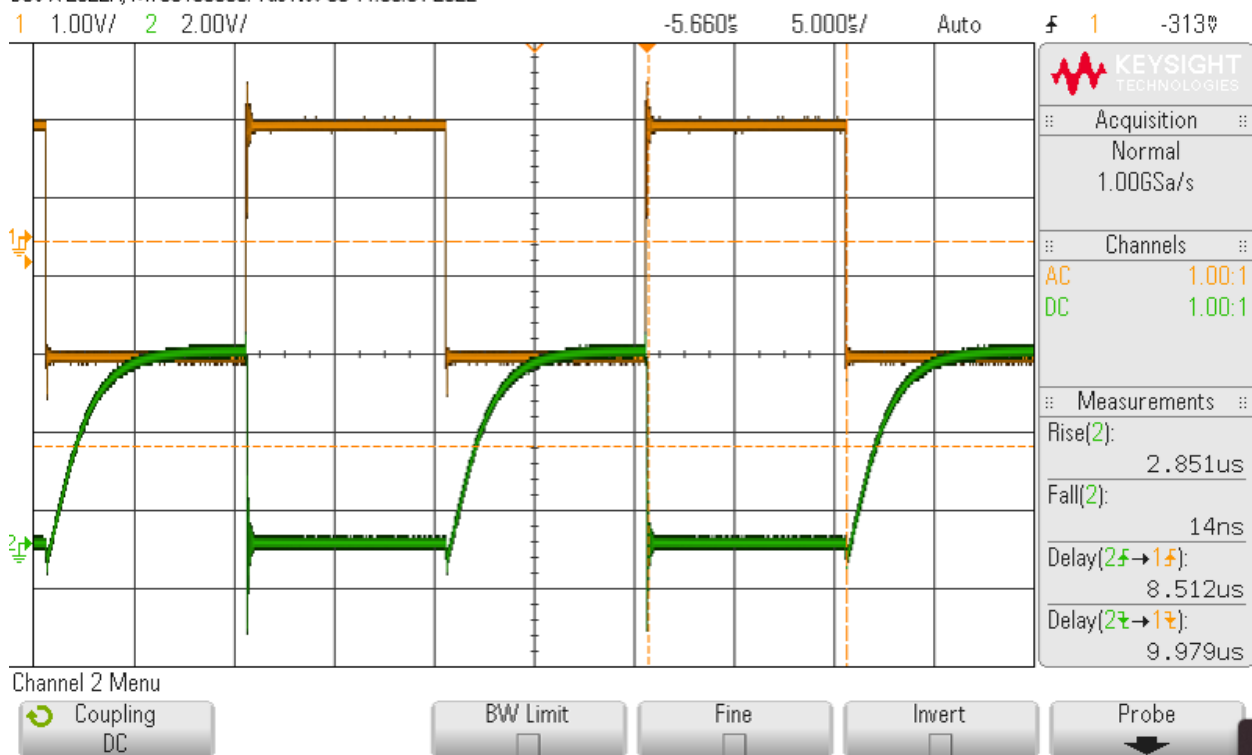
$R_c=0\Omega$, $R_o=1k\Omega$:

DSO-X 2022A, MY53160365: Tue Nov 08 11:49:26 2022



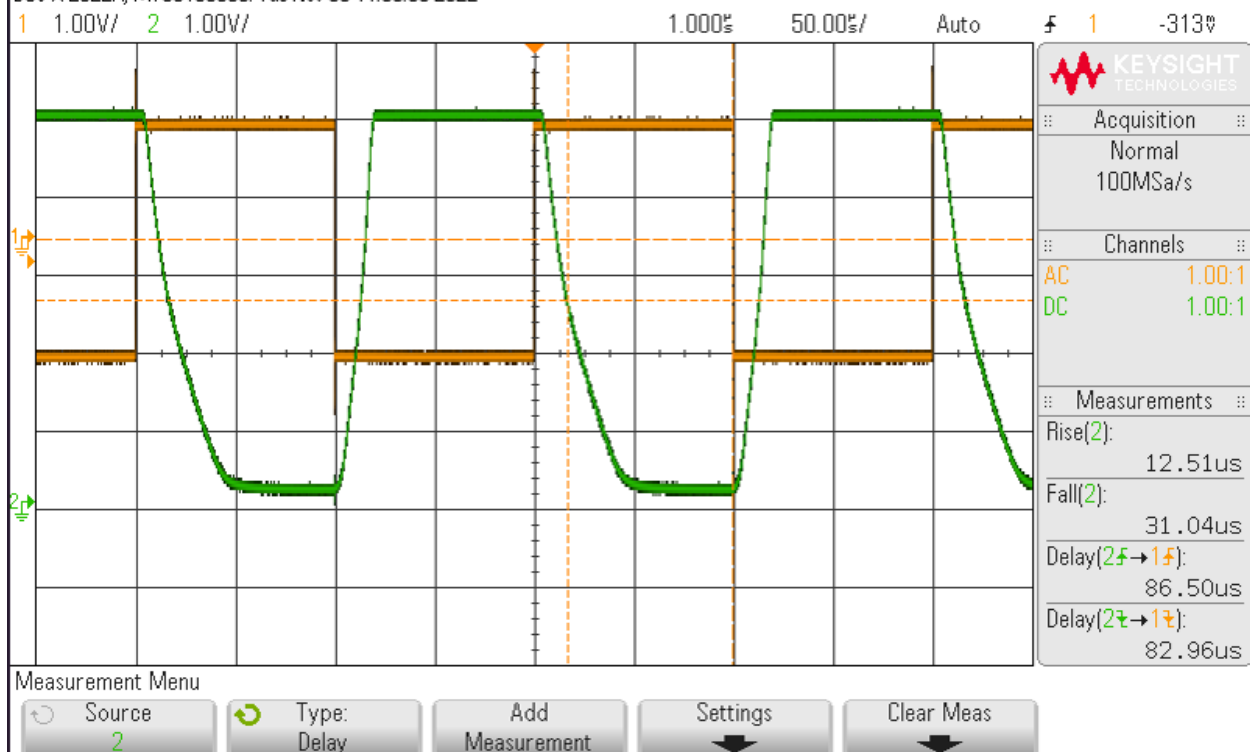
$R_c=0\Omega$, $R_o=10k\Omega$:

DSO-X 2022A, MY53160365: Tue Nov 08 11:50:01 2022



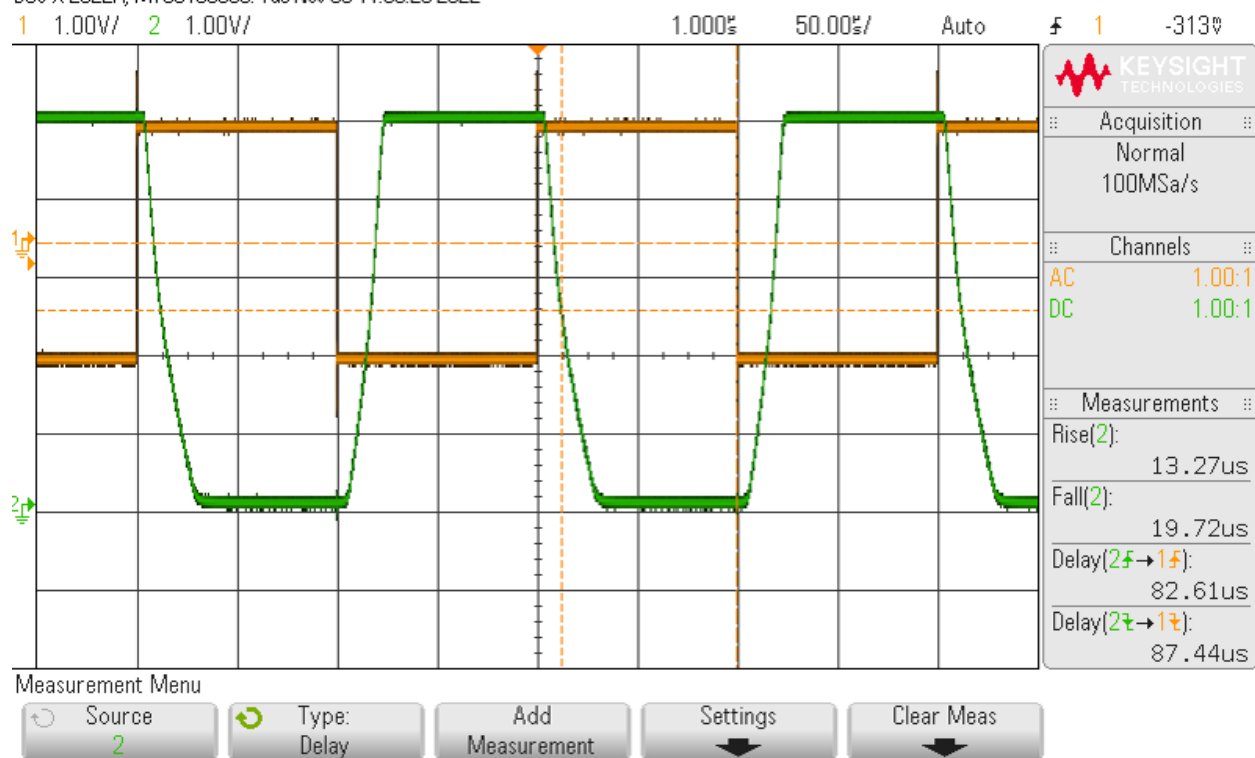
$R_c=100k\Omega$, $R_o=100\Omega$:

DSO-X 2022A, MY53160365: Tue Nov 08 11:56:08 2022



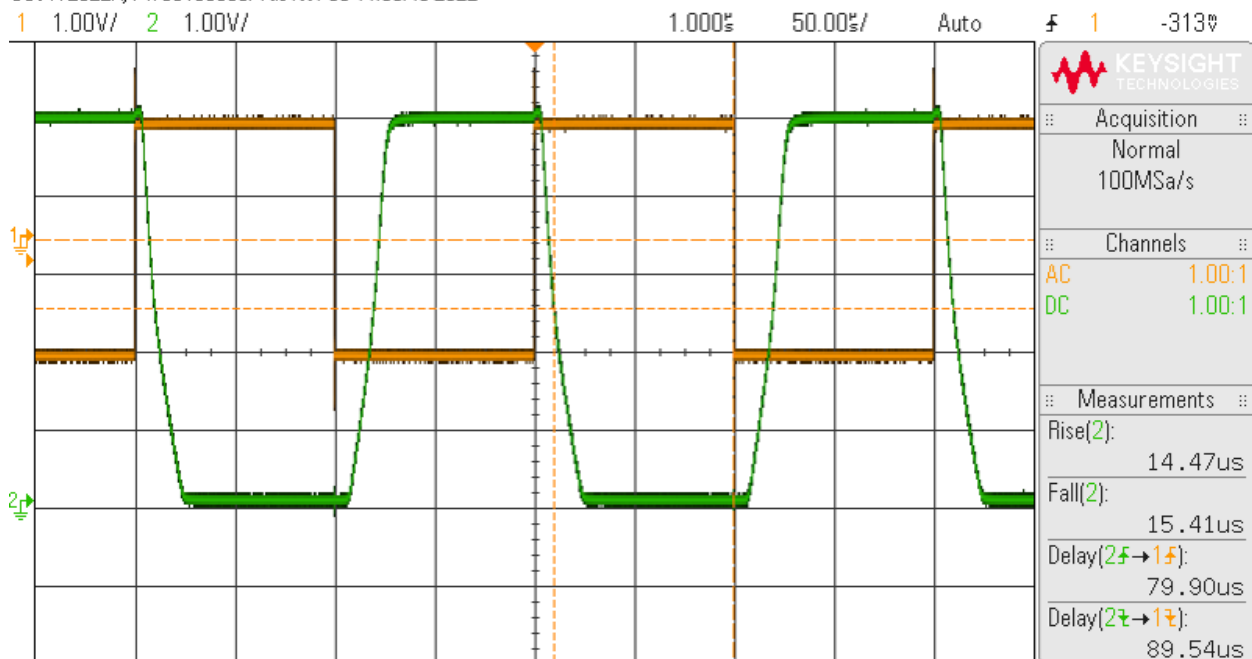
$R_o=100k\Omega$, $R_o=1k\Omega$:

DSO-X 2022A, MY53160365: Tue Nov 08 11:56:29 2022



$R_o=100k\Omega$, $R_o=10k\Omega$:

DSO-X 2022A, MY53160365: Tue Nov 08 11:56:49 2022



Measurement Menu

Source 2 Type: Delay Add Measurement Settings Clear Meas

Rise Time			
Rg\Rd	100Ω	1kΩ	10kΩ
0Ω	15ns	314ns	2.851μs
100kΩ	12.51μs	13.27μs	14.47μs

Fall Time			
Rg\Rd	100Ω	1kΩ	10kΩ
0Ω	15ns	14ns	14ns
100kΩ	31.04μs	19.72μs	15.41μs

Time Delay in the Rise			
Rg\Rd	100Ω	1kΩ	10kΩ
0Ω	0s	1μs	3.5μs
100kΩ	15μs	18μs	27μs

Time Delay in the Fall			
Rg\Rd	100Ω	1kΩ	10kΩ
0Ω	0s	1μs	3μs
100kΩ	25μs	15μs	12.5μs